Magnetism and Electromagnets

Magnets produce magnetic fields around them. A magnetic field is the region where magnetic materials or other magnets experience a force. Magnetic field lines are used to show the shape and direction of a magnetic field. Outside of a magnet, magnetic field lines always travel from the north pole to the south pole. The closer the field lines are together, the stronger the magnetic field is in that area. Magnetic field lines never cross each other.

Permanent magnets always produce a magnetic field. Examples include fridge magnets and bar magnets. Electromagnets are different because they only produce a magnetic field when an electric current flows through them. Electromagnets are made by wrapping a coil of wire around an iron core and passing a current through the wire. Electromagnets are useful because they can be switched on and off, and their strength can be changed.

Some magnetic materials can become temporary magnets when placed in a magnetic field. Materials such as iron, nickel, and cobalt can become magnetised when they are brought close to a magnet. This happens because the magnetic field causes tiny magnetic regions inside the material to line up in the same direction. Iron is particularly useful for this because it is easy to magnetise and demagnetise. When the magnetic field is removed, the material may lose most or all of its magnetism. This is why iron is commonly used as the core of electromagnets.

There are several ways to make an electromagnet stronger. Increasing the current in the wire will increase the strength of the magnetic field. Adding more turns of wire around the iron core will also make the electromagnet stronger. Using an iron core instead of an air core greatly increases the strength of the electromagnet.

Magnetic fields can also exert forces on wires carrying electric currents. When a current carrying wire is placed in a magnetic field, a force acts on the wire. This is known as the motor effect. The size of the force can be calculated using the equation:

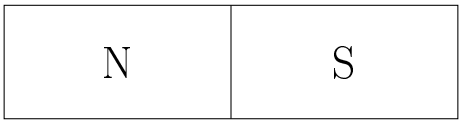
$$F = BIl$$

In this equation, F is the force in newtons (N), B is the magnetic flux density in tesla (T), I is the current in amps (A), and l is the length of the wire in metres (m).

Activity 1
Read the information above carefully. Write down any key points, important facts, formulas, or definitions into your exercise book.

Activity 2
Draw the magnetic field lines around each magnet arrangement below. Include arrows to show the direction of the field.

1. Single Bar Magnet



2. Two Magnets Attracting



3. Two Magnets Repelling



Activity 3
Match each keyword to its correct definition.

Magnetic field
Electromagnet
Permanent magnet
Solenoid
Magnetic field line
North pole
South pole
Motor effect

A magnet that only works when current flows
A coil of wire used to create a magnetic field
The region where magnetic forces act
The force produced on a current carrying wire in a magnetic field
The part of a magnet where field lines leave the magnet
A magnet that always produces a magnetic field
A line used to show the direction and shape of a magnetic field
The part of a magnet where field lines enter the magnet

Activity 4
Fill in the blanks below.

Electromagnets are useful because they can be switched _____ and _____.

The strength of an electromagnet can be changed by increasing the _____ in the wire.

Adding more _____ of wire around the core makes the electromagnet stronger.

Using an _____ core increases the strength of the electromagnet.

Permanent magnets always produce a magnetic field, but electromagnets only work when an electric _____ flows.

Activity 5
Use the equation below to answer the following questions.

$$F = BIl$$

1. Calculate the force when $B = 2\,T$, $I = 3\,A$, and $l = 0.5\,m$.
2. Calculate the force when $B = 1.5\,T$, $I = 4\,A$, and $l = 2\,m$.
3. Calculate the force when $B = 0.8\,T$, $I = 5\,A$, and $l = 0.4\,m$.
4. Calculate the force when $B = 3\,T$, $I = 2\,A$, and $l = 0.25\,m$.
5. Calculate the force when $B = 0.2\,T$, $I = 10\,A$, and $l = 1.5\,m$.
6. Calculate the current when $F = 6\,N$, $B = 2\,T$, and $l = 1\,m$.
7. Calculate the current when $F = 12\,N$, $B = 3\,T$, and $l = 2\,m$.
8. Calculate the magnetic flux density when $F = 10\,N$, $I = 5\,A$, and $l = 1\,m$.
9. Calculate the magnetic flux density when $F = 8\,N$, $I = 2\,A$, and $l = 2\,m$.
10. Calculate the wire length when $F = 9\,N$, $B = 3\,T$, and $I = 1.5\,A$.
11. Calculate the force when $B = 1.2\,T$, $I = 4\,A$, and $l = 30\,cm$.
12. Calculate the force when $B = 0.5\,T$, $I = 6\,A$, and $l = 20\,cm$.
13. Calculate the current when $F = 4\,N$, $B = 2\,T$, and $l = 40\,cm$.
14. Calculate the magnetic flux density when $F = 3\,N$, $I = 5\,A$, and $l = 15\,cm$.
15. Calculate the wire length when $F = 7.2\,N$, $B = 4\,T$, and $I = 3\,A$.

Answers

Activity 3

- Magnetic field → The region where magnetic forces act
- Electromagnet → A magnet that only works when current flows
- Permanent magnet → A magnet that always produces a magnetic field
- Solenoid → A coil of wire used to create a magnetic field
- Magnetic field line → A line used to show the direction and shape of a magnetic field
- North pole → The part of a magnet where field lines leave the magnet
- South pole → The part of a magnet where field lines enter the magnet
- Motor effect → The force produced on a current carrying wire in a magnetic field

Activity 4

- on / off
- current
- turns
- iron
- current

Activity 5

1. $3\,N$
2. $12\,N$
3. $1.6\,N$
4. $1.5\,N$
5. $3\,N$
6. $3\,A$
7. $2\,A$
8. $2\,T$
9. $2\,T$

10. $2\,m$
11. $1.44\,N$
12. $0.6\,N$
13. $5\,A$
14. $4\,T$
15. $0.6\,m$